Prothrombin Time (PT)

Test Details

METHODOLOGY

Prothrombin time reagent containing thromboplastin and calcium chloride is mixed with the patient plasma and the time to clot formation is measured photo-optically. The calcium chloride overcomes the citrate anticoagulant and allows the tissue factor in the thromboplastin to initiate coagulation. The prolongation of the clotting time correlates with the degree of deficiency or inhibition of the extrinsic pathway factors.

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

International normalized ratio (INR); prothrombin time

Use

Evaluation of the extrinsic coagulation system; aid in screening for congenital and acquired deficiencies of factors II, V, VII, X, and fibrinogen.[6-8] This test is used clinically for the therapeutic monitoring of warfarin (Coumadin®) anticoagulant therapy.

Special Instructions

Values obtained with different assay methods should not be used interchangeably in serial testing. It is recommended that only one assay method be used consistently to monitor each patient's course of therapy. This procedure does not provide serial monitoring; it is intended for one-time use only. If serial monitoring is required, please order the serial monitoring test, 485199. (/tests/485199/prothrombin-time-pt-serial-monitor)

If the patient's hematocrit exceeds 55%, the volume of citrate in the collection tube must be adjusted. Refer Coagulation Collection Procedures (/node/455) for directions.

Limitations

The PT test may not be sensitive to slight deficiencies of single factors.[9] Heparin can extend the PT.[9] Lupus anticoagulants may affect prothrombin time producing results that do not accurately reflect the true level of anticoagulation. The PT (and aPTT) results can be extended in patients receiving antithrombin anticoagulants such as hirudin and argatroban.[9] These patients can be monitored for warfarin effectiveness with a chromogenic factor X assay, which is not affected by lupus anticoagulant or thrombin inhibitors.[9]

Custom Additional Information

The thromboplastin reagent used by LabCorp consists of recombinant tissue factor mixed with synthetic phospholipid. Recombinant tissue factor is free from contamination with coagulation factors that can be found in tissue factor extracted from other sources. This serves to increase the PT assay's sensitivity for factor efficiencies. Prothrombin time results are reported in seconds and are also converted to international normalized ratio (INR) values. The INR serves to normalize results obtained from different laboratories for the variable responsiveness of different thromboplastin reagents. Each thromboplastin is assigned an activity Index (ISI) based on comparison to an international reference thromboplastin from the World Health Organization. The formula for calculating the INR is

$$\text{INR} = [\text{patient's results} / \text{normal patient average}]^{(\text{ISI})}$$

The ISI of the thromboplastin used in the LabCorp assay is near 1.0. The use of low ISI thromboplastin serves to improve the precision of therapeutic monitoring by enhancing sensitivity of the prothrombin assay.

The PT is sensitive to deficiencies of extrinsic and common pathway factors X, VII, V, II, and fibrinogen.[6-8] The PT is more responsive to deficiencies of factors X and V than is the aPTT. Congenital deficiencies of these factors are relatively rare and cause bleeding disorders of varying severity. Refer to individual test descriptions for more information. Acquired deficiencies of the vitamin K-dependent factors II and VII may occur during warfarin therapy and in patients with vitamin K deficiency. Diminished levels of all the factors of the extrinsic pathway can also occur in consumptive coagulopathies, such as disseminated intravascular coagulation (DIC), and as the result of decreased factor production as can be seen with severe liver disease or malnutrition. Specific inhibitors of extrinsic pathway factors are extremely rare, but may produce a prolonged PT.[6-8] Lupus anticoagulants (LA) may cause a prolonged PT due to nonspecific factor inhibition. Some individuals with LA can develop antibodies that bind to and increase the rate of clearance of prothrombin (factor II). These patients typically have an extended PT due to reduced factor II level and an increased risk of bleeding.

Coumarins, a family of compounds that inhibit the vitamin K-dependent carboxylation of several coagulation factors, are commonly used to limit fibrin clot formation in individuals with increased risk of venous or arterial thrombosis.[10-12] Warfarin, also referred to as Coumadin®, is the most commonly used coumarin in North America.[11] Overdosing with warfarin can increase the risk of hemorrhage and inadequate dosing decreases the efficacy of anticoagulation. Unfortunately, a large number of factors can affect the pharmacological potency of these oral anticoagulants. These factors are reviewed in considerable detail in the American Heart Association/American College of Cardiology Foundation Guide to Warfarin Therapy.[12] Therapeutic monitoring is essential to maintain the dosage within the appropriate range range. Because the PT is sensitive to deficiencies of vitamin K-dependent factors II and VII, it is used to monitor warfarin therapy.

Coumarins inhibit the carboxylation of procoagulant factors II, VII, IX, X, and anticoagulants proteins C and S to a similar extent. Steady-state levels of these proteins are all reduced to a similar degree, based on the dose and effectiveness of the oral anticoagulant given; however, during the initial days of treatment, the rate of decline of factor levels is dependent on their half-life. Since factor VII has a short half-life relative to other vitamin K-dependent factors, the levels of this factor drop much more precipitously than the others.[11]

Superwarfarins, such as brodifacoum, are often found in rat poison and may cause prolongation of the PT. Patients suffering brodifacoum poisoning respond to vitamin K often bringing their PT into the normal range briefly, but since the drug is stored for long periods of time in fat, the PT rises again over time.

Specimen Requirements

Specimen

Whole blood **or** plasma

Volume

2.7 mL-sized tube filled to 90% capacity **or** 1.8 mL-sized tube filled to 90% capacity.

Minimum Volume

90% of full draw (**Note:** This volume does **not** allow for repeat testing.)

Container

Blue-top (sodium citrate) tube; do **not** open tube unless plasma is to be frozen.

Collection Instructions

Blood should be collected in a blue-top tube containing 3.2% buffered sodium citrate.[1] Evacuated collection tubes must be filled to completion to ensure a proper blood-to-anticoagulant ratio.[2,3] The sample should be mixed immediately by gentle inversion at least six times to ensure adequate mixing of the anticoagulant with the blood. A discard tube is not required prior to collection of coagulation samples unless the sample is collected using a winged (butterfly) collection system. With a winged blood collection set a discard tube should be drawn first to account for the dead space of the tubing and prevent under-filling of the evacuated tube.[4,5] When noncitrate tubes are collected for other tests, collect sterile and nonadditive (red-top) tubes prior to citrate (blue-top) tubes. Any tube containing an alternative anticoagulant should be collected after the blue-top tube. Gel-barrier tubes and serum tubes with clot initiators should also be collected after the citrate tubes.

Please print and use the *Volume Guide for Coagulation Testing* *(/tests/related-documents/L5901)* to ensure proper draw volume.

Reference Range

See table.

Age	Range
Prothrombin Time (PT)	
0 to 30 d	9.8−13.2 sec
1 to 6 m	9.8−12.6 sec
7 m to 1 y	9.8−12.2 sec
2 to 17 y	9.9–12.1 sec
≥18 y	9.1−12.0 sec
International Normalized Ratio (INR)	
0 to 1 m	0.9−1.4
>1 m	0.9−1.2

Storage Instructions

Specimens are stable at room temperature for 24 hours. If testing cannot be completed within 24 hours, specimens should be centrifuged for at least 10 minutes at 1500xg; plasma should then be transferred to a Labcorp PP transpak frozen purple tube with screw cap (Labcorp No. 49482). **Freeze** immediately and maintain frozen until tested. Refer to Coagulation Collection Procedures (/node/455) for directions.

Causes for Rejection

Clotted specimen; hemolysis; tube <90% full; improper labeling; specimen collected in tube other than 3.2% citrate; refrigerated specimen

References

Lippi G, Adcock D, Simundic AM, Tripodi A, Favaloro EJ. Critical laboratory values in hemostasis: toward consensus. *Ann Med*. 2017 Sep; 49(6):455-461. PubMed 28042729 (http://www.ncbi.nlm.nih.gov/pubmed/28042729)

Footnotes

1. Adcock DM, Kressin DC, Marlar RA. Effect of 3.2% vs 3.8% sodium citrate concentration on routine coagulation testing. *Am J Clin Pathol.* 1997Jan; 107(1):105-110. PubMed 8980376 (http://www.ncbi.nlm.nih.gov/pubmed/8980376)
2. Reneke J, Etzell J, Leslie S, Ng VL, Gottfried EL. Prolonged prothrombin time and activated partial thromboplastin time due to underfilled specimen tubes with 109 mmol/L (3.2%) citrate anticoagulant. *Am J Clin Pathol.* 1998 Jun; 109(6):754-757. PubMed 9620035 (http://www.ncbi.nlm.nih.gov/pubmed/9620035)
3. National Committee for Clinical Laboratory Standardization. *Collection, Transport, and Processing of Blood Specimens for Coagulation Testing and General Performance of Coagulation Assays; Approved Guideline.* 5th ed. Villanova, Pa: NCCLS; 2008. Document H21-A5:28(5).
4. Gottfried EL, Adachi MM. Prothrombin time and activated partial thromboplastin time can be performed on the first tube. *Am J Clin Pathol.* 1997 Jun; 107(6):681-683. PubMed 9169665 (http://www.ncbi.nlm.nih.gov/pubmed/9169665)
5. McGlasson DL, More L, Best HA, Norris WL, Doe RH, Ray H. Drawing specimens for coagulation testing: Is a second tube necessary? *Clin Lab Sci.* 1999 May-Jun; 12(3):137-139. PubMed 10539100 (http://www.ncbi.nlm.nih.gov/pubmed/10539100)
6. Adcock DM, Bethel MA, Macy PA, et al. *Coagulation Handbook.* Esoterix Coagulation; 2006.
7. Roberts HR, Escobar MA. Less common congenital disorders of hemostasis. In Kitchens CS, Alving BM, Kessler CM, eds. *Consultative Hemostasis and Thrombosis.* Philadelphia, Pa: WB Saunders Co; 2002: 57-71.

8. Triplett DA. Coagulation abnormalities. In McClatchey KD, ed. *Clinical Laboratory Medicine.* 2nd ed. Philadelphia, Pa: Lippincott Williams and Wilkins; 2002:1033-1049.

9. Van Cott EM, Laposata M. Coagulation. In Jacobs DS, DeMott WR, Oxley DK, eds. *Laboratory Test Handbook With Key Word Index.* Hudson, Ohio: Lexi-Comp; 2001: 327-358.

10. Hirsh J, Dalen J, Anderson DR, et al. Oral anticoagulants: Mechanism of action, clinical effectiveness, and optimal therapeutic range. *Chest.* 2001 Jan; 119(1 Suppl):8S-21S. PubMed 11157640 (http://www.ncbi.nlm.nih.gov/pubmed/11157640)

11. Francis CW, Berkowitz SD. Antithrombotic and thrombolytic agents. In Kitchens CS, Alving BM, Kessler CM, eds. *Consultative Hemostasis and Thrombosis.* Philadelphia, Pa: WB Saunders Co; 2002: 375-391.

12. Hirsh J, Fuster V, Ansell J, Halperin JL; American Heart Association; American College of Cardiology Foundation. American Heart Association/American College of Cardiology Foundation guide to warfarin therapy. *Circulation.* 2003 Apr 1; 107(12):1692-1711. PubMed 12668507 (http://www.ncbi.nlm.nih.gov/pubmed/12668507)